

FMPT-6000 Smart Gauge / Absolute Pressure Transmitter

The FMPT-6000 is a smart Gauge and Absolute Pressure Transmitter built on monocrystalline silicon piezoresistive sensor technology. It measures the level, density, and pressure of liquid, gas, or steam and converts the pressure signal into a 4-20 mA DC analog current output, with configuration, monitoring, and diagnostics available through HART or Modbus-RTU communication — suited to demanding process and chemical-industry duty.

General Features

- Monocrystalline silicon piezoresistive sensor technology with stable performance, accuracy up to $\pm 0.075\%$ FS.
- High overload protection diaphragm for robust pressure overload protection.
- High-performance EMC protection circuit module with strong anti-interference ability.
- Long-term stability: $\pm 0.2\%$ SPAN / 5 years.
- Output via 4-20 mA DC + HART, or Modbus-RTU / RS-485.



Key Specifications

Measurement	Gauge (GP) & Absolute (AP)
Range	GP 0.4-400 bar; AP 0.4-100 bar
Accuracy	GP $\pm 0.075\%$; AP $\pm 0.1\%$ URL
Turn-Down	100:1
Output	4-20 mA HART, or Modbus-RTU
Stability	$\pm 0.2\%$ Span / 5 years
IP Rating	IP67

Industries & Applications

- Oil and Gas industry
- Food processing
- Pulp and Paper
- Power and Energy
- Chemical Industry
- Marine Equipment

General Specifications — FMPT-6000

Parameter	Specification
Measurement Type	Gauge Pressure (GP) and Absolute Pressure (AP)
Nominal Range	GP: 0.4 bar to 400 bar (6 ranges) AP: 0.4 bar to 100 bar (4 ranges)
Turn-Down Ratio	100:1
Accuracy (Linear)	GP: $\pm 0.075\%$ to $\pm 0.2\%$ URL AP: $\pm 0.1\%$ to $\pm 0.2\%$ URL
Long-Term Stability	$\pm 0.2\%$ Span / 5 years
Output Signal	4-20 mA DC + HART, or Modbus-RTU / RS-485
Sensor Technology	Monocrystalline silicon piezoresistive
Power Supply Voltage	HART: 16.5-55 V DC; IS HART: 18.5-28 V DC; Modbus: 12-32 V DC
Power Consumption	4-20 mA: ≤ 500 mW @ 24 V DC, 20.8 mA; Modbus: ≤ 240 mW @ 24 V DC, 10 mA
Load Resistance	0-2119 Ohm (operating); 250-600 Ohm (HART communication)
Damping (Electronic)	0-100 s configurable; sensor module damping ≤ 0.2 s
Certification	Contact Forbes Marshall for current hazardous-area and regulatory approval status
IP Rating	IP67
Weight	About 1.56 kg (excluding mounting bracket and process connection parts)

Measurement Range Specifications

Type	Nominal Range	Min. Range	Lower (LRL)	Upper (URL)	Overload
GP	0.4 bar	20 mbar	-0.4 bar	0.4 bar	10 bar
GP	2.5 bar	125 mbar	-1 bar	2.5 bar	40 bar
GP	10 bar	0.5 bar	-1 bar	10 bar	60 bar
GP	30 bar	1.5 bar	-1 bar	30 bar	150 bar
GP	100 bar	5 bar	-1 bar	100 bar	200 bar
GP	400 bar	50 bar	-1 bar	400 bar	800 bar
AP	0.4 bar	0.2 bar	0 bar	0.4 bar	10 bar
AP	2.5 bar	0.5 bar	0 bar	2.5 bar	40 bar
AP	10 bar	2 bar	0 bar	10 bar	60 bar
AP	100 bar	10 bar	0 bar	100 bar	200 bar

Note: AP nominal range can also be selected as 30 bar; consult Forbes Marshall for details.

Accuracy — Linear Output

Type	Condition	Range	Accuracy
GP	TD ≤ 5	0.4 bar	±0.1%
GP	TD ≤ 5	2.5, 10, 30, 100, 400 bar	±0.075%
GP	TD > 5	0.4 bar	±(0.025 + 0.015 × TD)%
GP	TD > 5	2.5, 10, 30, 100, 400 bar	±(0.0025 + 0.0145 × TD)%
AP	TD ≤ 5	0.4 bar	±0.2%
AP	TD ≤ 5	2.5, 10, 100 bar	±0.1%
AP	TD > 5	0.4 bar	±(0.025 + 0.035 × TD)%
AP	TD > 5	2.5, 10, 100 bar	±(0.025 + 0.015 × TD)%

Note: TD (Turn-Down) = URL (maximum/factory-calibrated range) / Current SPAN. Gold-plated 316L diaphragm at 0.4 bar can reach 0.1% FS.
Reference: 20°C ±5°C, calibrated from zero.

Ambient Temperature Effects

Type	Effect	Applicable Range
GP	±(0.075 + 0.035 × TD)% per 10°C of SPAN	0.4, 2.5, 10, 30, 100, 400 bar
AP	±(0.125 + 0.075 × TD)% per 10°C of SPAN	0.4 bar
AP	±(0.115 + 0.065 × TD)% per 10°C of SPAN	2.5, 10, 100 bar

Environmental Conditions

Parameter	Specification
Working Temperature	-40 to +85°C; With LCD: -20 to +70°C
Storage Temperature	-40 to +100°C; With LCD: -40 to +85°C
Media Temperature	Silicone oil filled: -40 to +105°C; Inert oil filled: -45 to +85°C
Working Humidity	5% RH to 100% RH @ 40°C
Power Supply Effect	For 10.5V/16.5V to 55V DC supply, zero/span change ≤0.005% URL/V
Mounting Position Effects	< 4 mbar at any position, correctable by PV = 0 reset
Vibration Effect	< 0.1% URL per GB/T18271.3 / IEC 61298-3

EMC Environment

Test	Standard	Test Condition	Level
Radiated Interference (Casing)	GB/T 9254.1 / CISPR 32	30 MHz - 1000 MHz	Qualified
Conducted Interference (DC Port)	GB/T 9254.1 / CISPR 32	0.15 MHz - 30 MHz	Qualified
Electrostatic Discharge Immunity	GB/T 17626.2 / IEC 61000-4-2	8 kV contact / 8 kV air	A
Radiated RF/EM Field Immunity	GB/T 17626.3 / IEC 61000-4-3	10 V/m (80 MHz - 1 GHz)	A
Power Frequency Magnetic Field	GB/T 17626.8 / IEC 61000-4-8	30 A/m	A
Electrical Fast Transient / Burst	GB/T 17626.4 / IEC 61000-4-4	4 kV (5/50 ns, 100 kHz)	A
Surge Immunity	GB/T 17626.5 / IEC 61000-4-5	1 kV (L-L); 2 kV (L-G)	A
RF Conducted Disturbance Immunity	GB/T 17626.6 / IEC 61000-4-6	3 V/m (150 kHz - 80 MHz)	A

Note: Level A: performance is normal within the limits of the technical specifications.

Power Supply, Load & Time Index

Parameter	Specification
Power Supply Voltage	HART: 16.5-55 V DC IS HART: 18.5-28 V DC Modbus-RTU/RS-485: 12-32 V DC
Load Resistance	0-2119 Ohm (working state); 250-600 Ohm (HART communication)
Transmission Distance	< 1000 m
Power Consumption	4-20 mA: ≤500 mW @ 24 V DC, 20.8 mA Modbus: ≤240 mW @ 24 V DC, 10 mA
Damping (Electronic)	0-100 s configurable
Sensor Module Damping	≤ 0.2 s
Turn-On Time	≤ 6 s
Factory Reset Time	≤ 31 s

Sensor & Wetted Parts Options

Parameter	Code	Description
Isolated Diaphragm Material	S / H / L	S=SS316L; H=Hastelloy C; L=SUS316L + gold plated
Fill Oil	S / D	S=Silicone oil; D=Inert filler
Sensor Seal	F / S	F=Stainless steel welded seal; S=Viton O-ring

Electrical Connection (Cable Outlet Protection)

Code	Description	Material	IP Rating
R1	Waterproof connector M20x1.5 + gland, opposite side	PVC	IP67 (6-8mm cable)
R2	Flameproof, 1/2 NPT (F) + gland, opposite side	Stainless steel	IP67 (6-8mm cable)
R3	Flameproof, M20x1.5 (F) + gland, opposite side	Stainless steel	IP67 (6-8mm cable)
R7	Flameproof, G1/2 (F) + gland, opposite side	Stainless steel	IP67 (6-8mm cable)

Output Signal & Display

Code	Description
H	4-20 mA DC + HART, two-wire, power supply 16.5-55 V DC
B	4-20 mA DC + HART, two-wire, intrinsically safe, power supply 18.5-28 V DC
R	Modbus-RTU / RS-485, applicable power supply 12-32 V DC
Display A / C	A = without LCD display; C = with LCD display

Process Connection Options

Code	Specification
M01 / M10	M20 x 1.5 Male (Phi 3 mm vent) / M20 x 1.5 Female
G01 / G02	G1/2 Male (Phi 3 mm vent) / G1/4 Male (Phi 3 mm vent)
R01 / R03	1/2-14 NPT Male (Phi 6 mm) / 1/2-14 NPT Female (Phi 6 mm)
R02 / R04	1/4-18 NPT Male (Phi 3 mm) / 1/4-18 NPT Female (Phi 3 mm)
R35	1/2-14 NPT Male (includes 1/4-18 NPT)

Ordering Information — Model Code Structure

The FMPT-6000 model code is structured as follows. Please specify all parameters when placing an order. Factory calibration range (LRL-URL) and display unit must be stated in the purchase contract.

Block	Field	Key Options
Model	Product	FMPT-6000-GP or FMPT-6000-AP
Sensor	Range	GP: 0.4/2.5/10/30/100/400 bar; AP: 0.4/2.5/10/100 bar (see range table)

Block	Field	Key Options
Sensor	Diaphragm / Fill Oil / Seal	S/H/L diaphragm; S/D fill oil; F/S seal
Electrical Connection	Cable Outlet	R1=Waterproof M20; R2=FP 1/2NPT; R3=FP M20; R7=FP G1/2
Output	Signal / Display	H=HART; B=IS HART; R=Modbus-RTU; A=No LCD; C=LCD
Process Connection	Specification	M01/M10/G01/G02/R01-R04/R35 (see table above)
/Options	Fittings & Mounting	/N1,/N2=Heat exchange link; /Z1,/Z2=Welded connector; /B4,/B7=U-bracket
/Options	Other	/Q1=Validation report; /G1=Ungrease treatment

Code format: the Sensor block (Range + Diaphragm + Fill Oil + Seal) is written as one unbroken code, and the Output block (Signal + Display) is written as one unbroken code; a single dash separates each major block.

Example: FMPT-6000-GP-S105GSSF-R2-HC-M01/B7 [CAL: 0 to 10 bar, bar display]

Dimensional Reference (mm)

Parameter	With Display	Without Display
Overall Width	131 mm (93 mm body)	131 mm (93 mm body)
Overall Depth	133 mm	110 mm
Overall Height (upper)	154 mm	154 mm
Overall Height (lower, to port)	119 mm	119 mm
Process Connection Thread	M20 x 1.5 (base configuration)	
U-Bracket (2" pipe mount, /B4, /B7)	176 x 160 mm envelope	

Note: Product is factory-assembled with NPT plug and drain valve. Actual dimensions vary with process adapter and mounting bracket selection.

Quick Reference — Common Configurations

Worked examples for reference only; confirm every field against the full model code table before ordering.

Configuration	Order Code	Range
Standard GP, SS316L, silicone oil, HART, LCD	FMPT-6000-GP-S105GSSF-R1-HC-M01	0-10 bar
AP, Hastelloy diaphragm, Modbus, no LCD, NPT	FMPT-6000-AP-S105AHDS-R2-RA-R01	0-10 bar